

Introduction to MongoDB

What is a Database?

What is a Database?

A **database** is a place where we store data in an organized way so we can create, **read**, **update**, and **delete** it easily.

Real examples:

- Student records
- Product details in an e-commerce site

Without a database → data becomes messy, hard to manage.

MYSQL Vs MONGODB

Feature	MySQL	MongoDB
Database Type	Relational Database (RDBMS)	NoSQL Document Database
Data Storage	Tables	Collections
Record	Row	Document
Column	Column	Field (Key : Value)
Schema	Fixed Schema	Flexible Schema
Data Format	Rows & Columns	JSON/BSON Documents
Relationships	Foreign Key ದ್ವಾರಾ	Embedded Documents ಲೇದಾ References
Scalability	Vertical Scaling	Horizontal Scaling
Query Language	SQL	MongoDB Query Language (MQL)
Best For	Banking, ERP, College Manager 	Social Media, E-commerce, Chat Applications

Types of Databases

Two Types of Databases

Two Types of Databases

SQL Database

id	name	age
1	atridatta	37
2	balaji	35
3	srinu	7

Data stored in tables (rows & columns)

NoSQL Database

```
{  
  "id": 1,  
  "name": "atridatta",  
  "age": 37  
}
```

Data stored in documents (JSON)

Tables vs Collections



MongoDB ecosystem Tools



mongod (MongoDB Server)



mongosh (Mongo Shell)



MongoDB Compass



MongoDB Atlas

mongod and mongosh



mongod (MongoDB Server)

- This is the **database server**
It runs in background
- Stores all data
- Default port: 27017

⚠ Without mongod -- MongoDB does not run

```
show dbs
use college
db.faculty.insertOne(
  (name: "AtriDatta", branch: "CSE"))
db.faculty.find()
```



mongosh (Mongo Shell)

- Command line tool to talk to MongoDB
- We write commands here

```
show dbs
use college
db.faculty.insertOne(
  (name: "AtriDatta",
    branch: "CSE"))
db.faculty.find()
```

MongoDB Compass and Atlas



MongoDB Compass

- GUI (graphical tool)
- See databases visually
- Insert/edit documents without commands



MongoDB Atlas

- MongoDB on the cloud
- No installation needed
- Used in real projects

MongoDb Data Format

mongoDB → Document-Oriented No-SQL Database

Data

```
{  
  "id" : 1,  
  "name" : "atridatta",  
  "age" : 37,  
  "phno" : 9876543210  
}
```

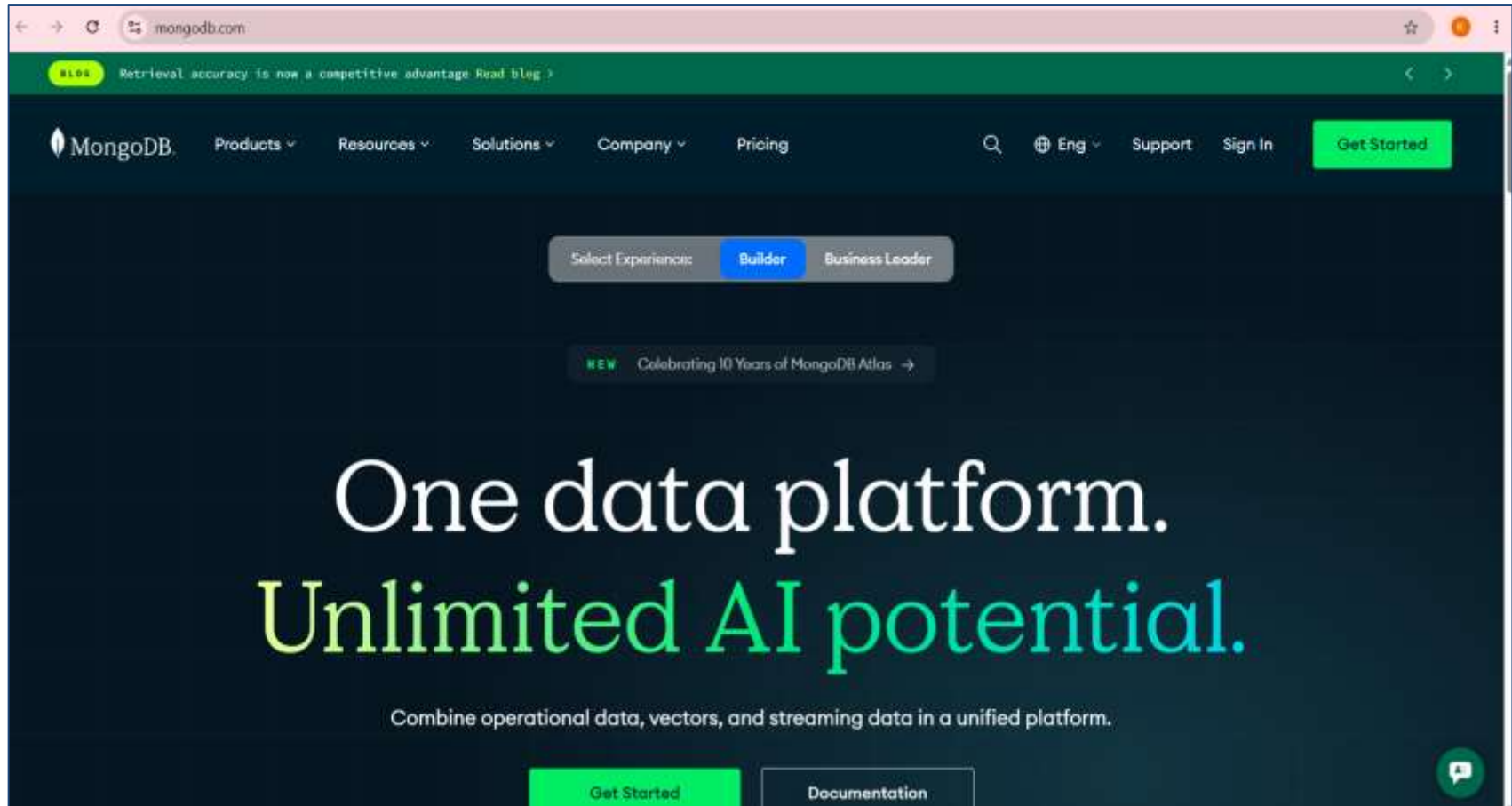
JSON STRUCTURE

MongoDB Installation

Step 1: Download MongoDB Community Server

Download **MongoDB Community Server** for Windows.

Open <https://www.mongodb.com/>



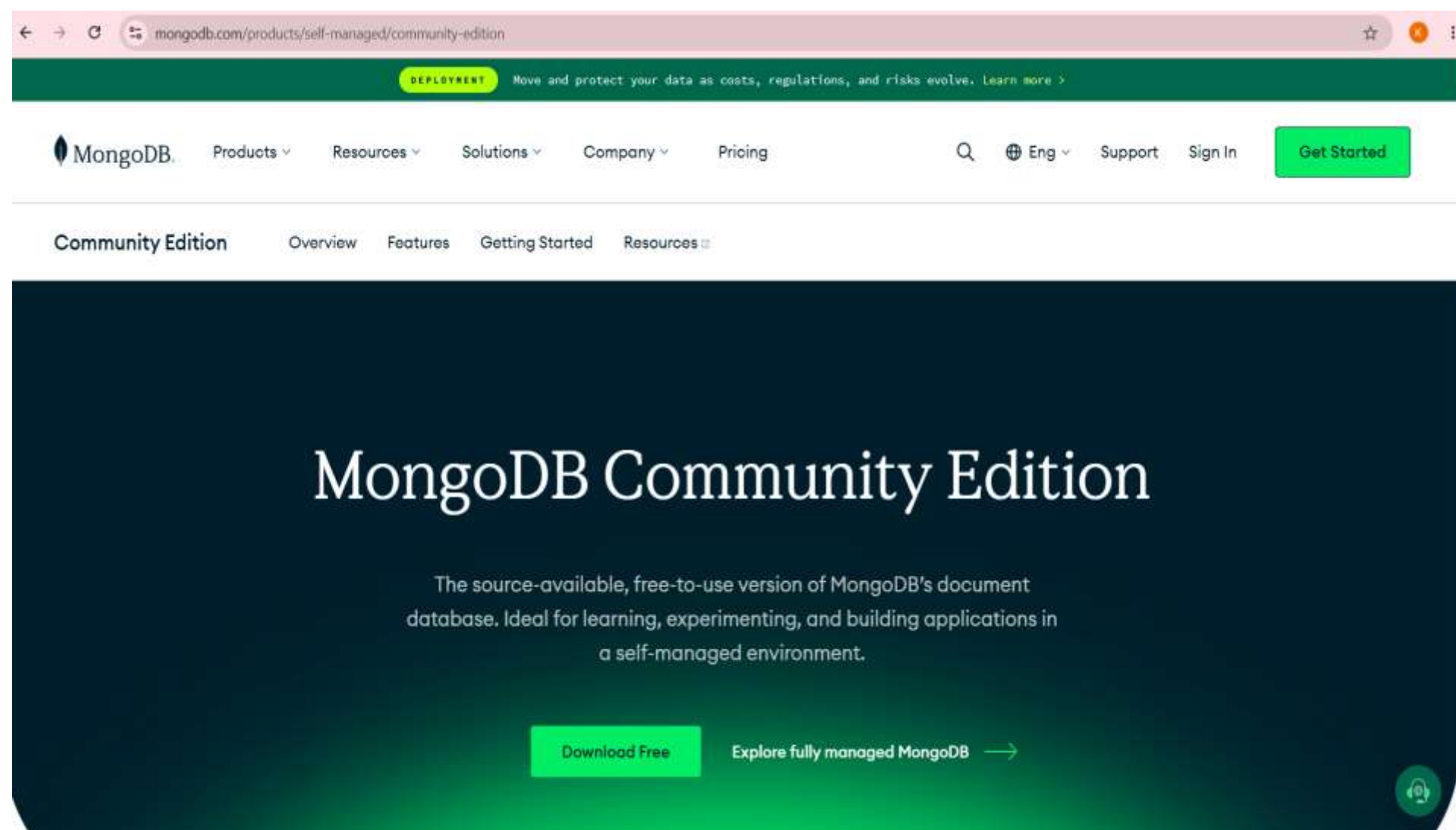
Goto Products Menu and Choose Community Edition

The screenshot shows the MongoDB website with the following elements:

- Header:** MongoDB logo, navigation links (Products, Resources, Solutions, Company, Pricing), search icon, language selector (Eng), Support, Sign In, and a prominent **Get Started** button.
- Platform Services:** A list of services including Atlas, Database, Search, Vector Search, and Stream Processing.
- Self-Managed:** A section containing Enterprise Advanced and Community Edition. The **Community Edition** option is highlighted with a dark background and white text, with the description "Develop locally with MongoDB".
- Tools:** A section containing Compass, Integrations, and Relational Migrator.
- Right Sidebar:** Three promotional cards for MongoDB Atlas, Enterprise Advanced, and Community Edition, each with a "Sign Up" or "Download" button.
- Footer:** A **Get Started** button and a **Documentation** link.

The URL in the browser address bar is <https://www.mongodb.com/products/self-managed/community-edition>.

Choose Download Free Button



The screenshot shows the MongoDB website's product page for the Community Edition. At the top, a green banner features a yellow 'DEPLOYMENT' button and a message about data protection. Below this is a navigation bar with the MongoDB logo, links for Products, Resources, Solutions, Company, and Pricing, a search icon, language selection (Eng), Support, Sign In, and a prominent green 'Get Started' button. The main content area has a dark blue background with the heading 'MongoDB Community Edition'. Below the heading, a paragraph describes it as a source-available, free-to-use version of MongoDB's document database, ideal for learning and experimentation. At the bottom, there are two buttons: a green 'Download Free' button and a blue 'Explore fully managed MongoDB' button with a right-pointing arrow. A small chat icon is visible in the bottom right corner.

← → ↻ ⓘ mongodb.com/products/self-managed/community-edition ☆ 🔔

DEPLOYMENT Move and protect your data as costs, regulations, and risks evolve. [Learn more >](#)

MongoDB Products ▾ Resources ▾ Solutions ▾ Company ▾ Pricing 🔍 🌐 Eng ▾ Support Sign In [Get Started](#)

Community Edition Overview Features Getting Started Resources ☰

MongoDB Community Edition

The source-available, free-to-use version of MongoDB's document database. Ideal for learning, experimenting, and building applications in a self-managed environment.

[Download Free](#) [Explore fully managed MongoDB →](#)

💬

Choose package as .msi and press download button

The screenshot shows the MongoDB download page for the community edition. The left sidebar lists various MongoDB products, with 'MongoDB Community Server' selected. The main content area displays three dropdown menus: 'Version' set to '8.3.4 (current)', 'Platform' set to 'Windows x64', and 'Package' set to 'msi'. Below these selections are three buttons: 'Download' (green), 'Copy link' (dark grey), and 'More Options' (grey). The 'Download' button is highlighted with a green border. The browser's address bar shows the URL 'mongodb.com/try/download/community'.

MongoDB Atlas

MongoDB Enterprise Advanced

MongoDB Community Edition

MongoDB Community Server

MongoDB Controllers for Kubernetes Operator

MongoDB Search in Community

Tools

SQL Interface

no downloads necessary.

Version
8.3.4 (current)

Platform
Windows x64

Package
msi

Download

Copy link

More Options

Windows Installer is downloaded

✓ Windows Installer Package

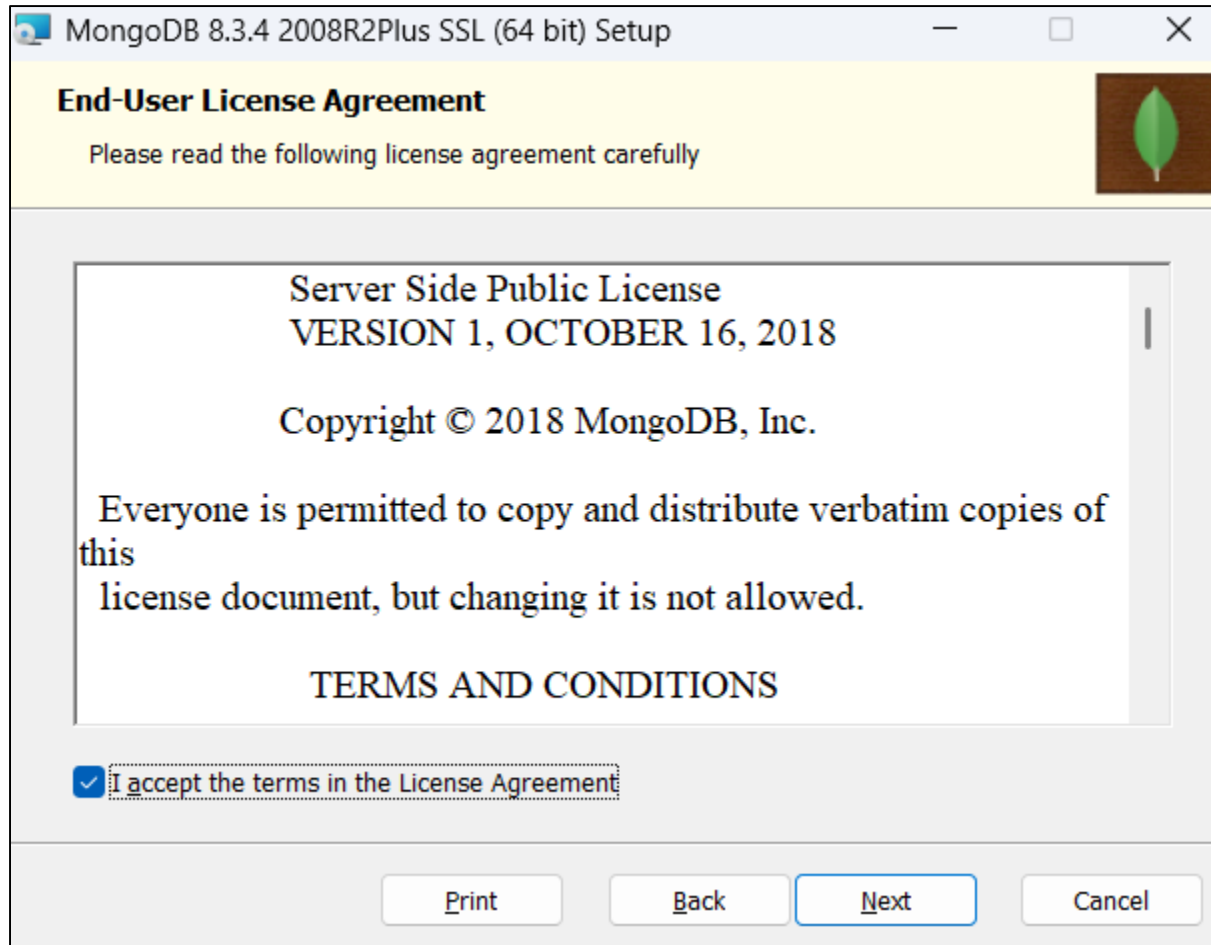


58 MB Available on this device

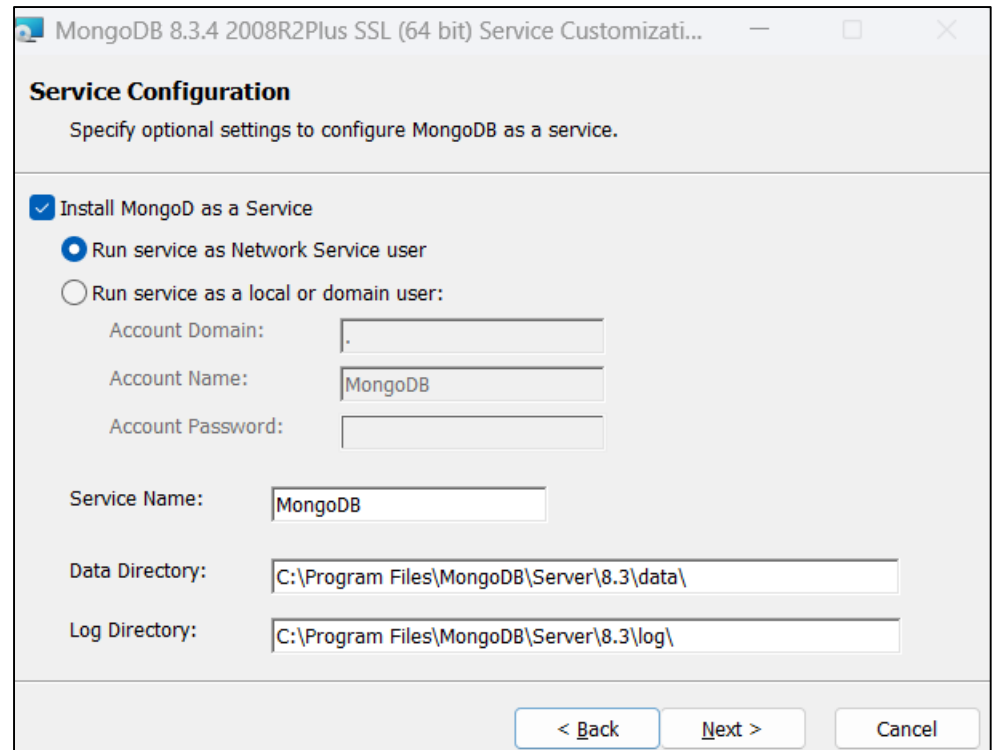
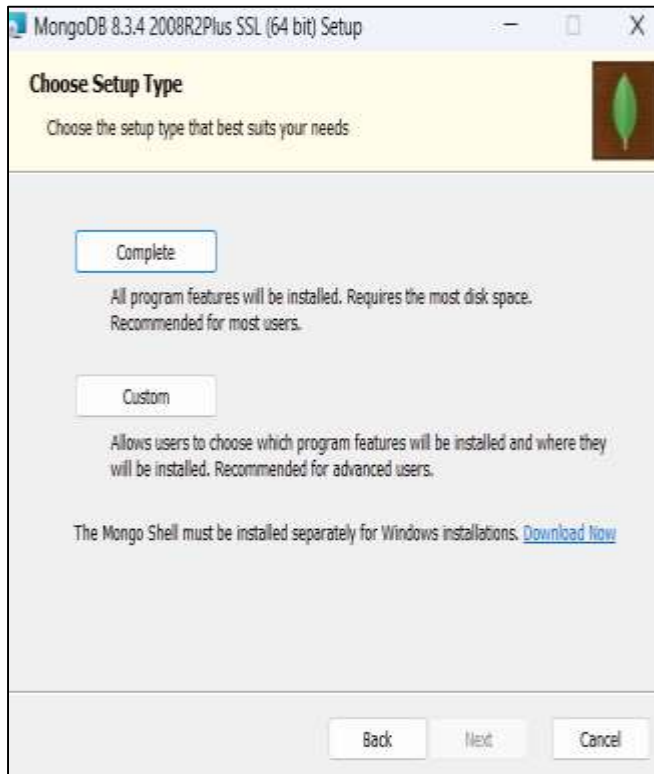
Now double click on that installer and press Next Button



Accept the License Agreement

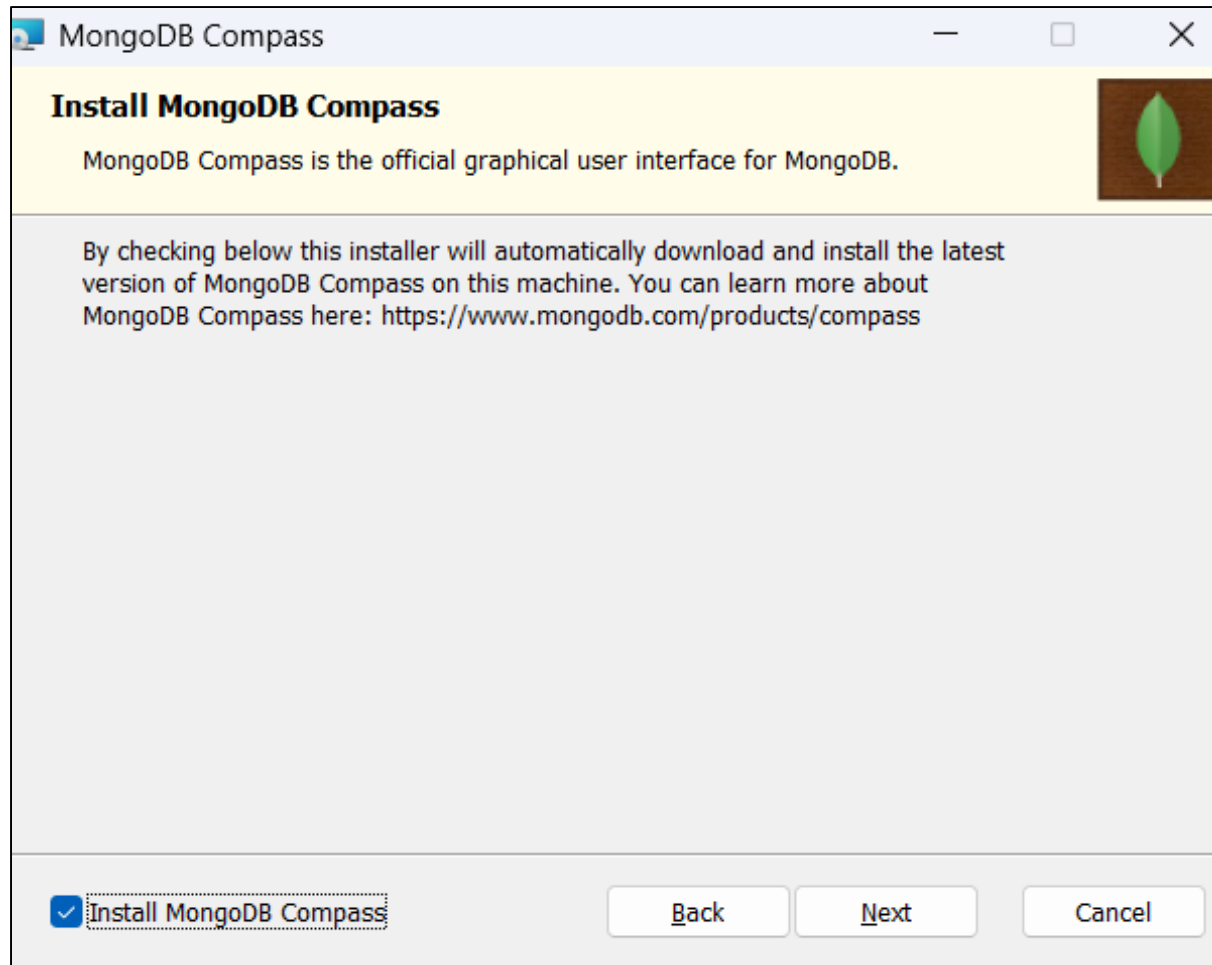


Select Complete Installation

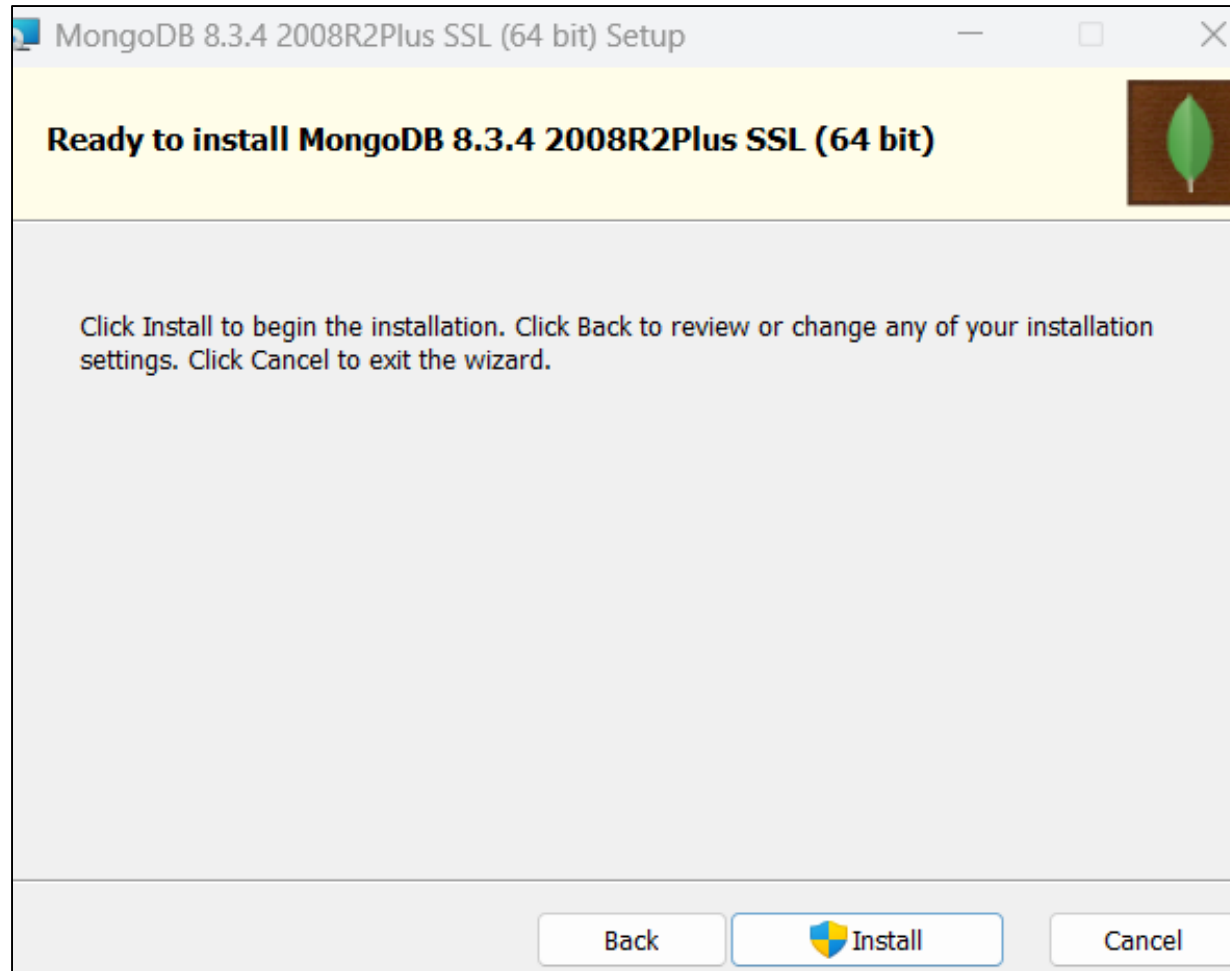


Click on Complete Installation and Choose Radio Button Run Service as Network Service User and press NEXT Button

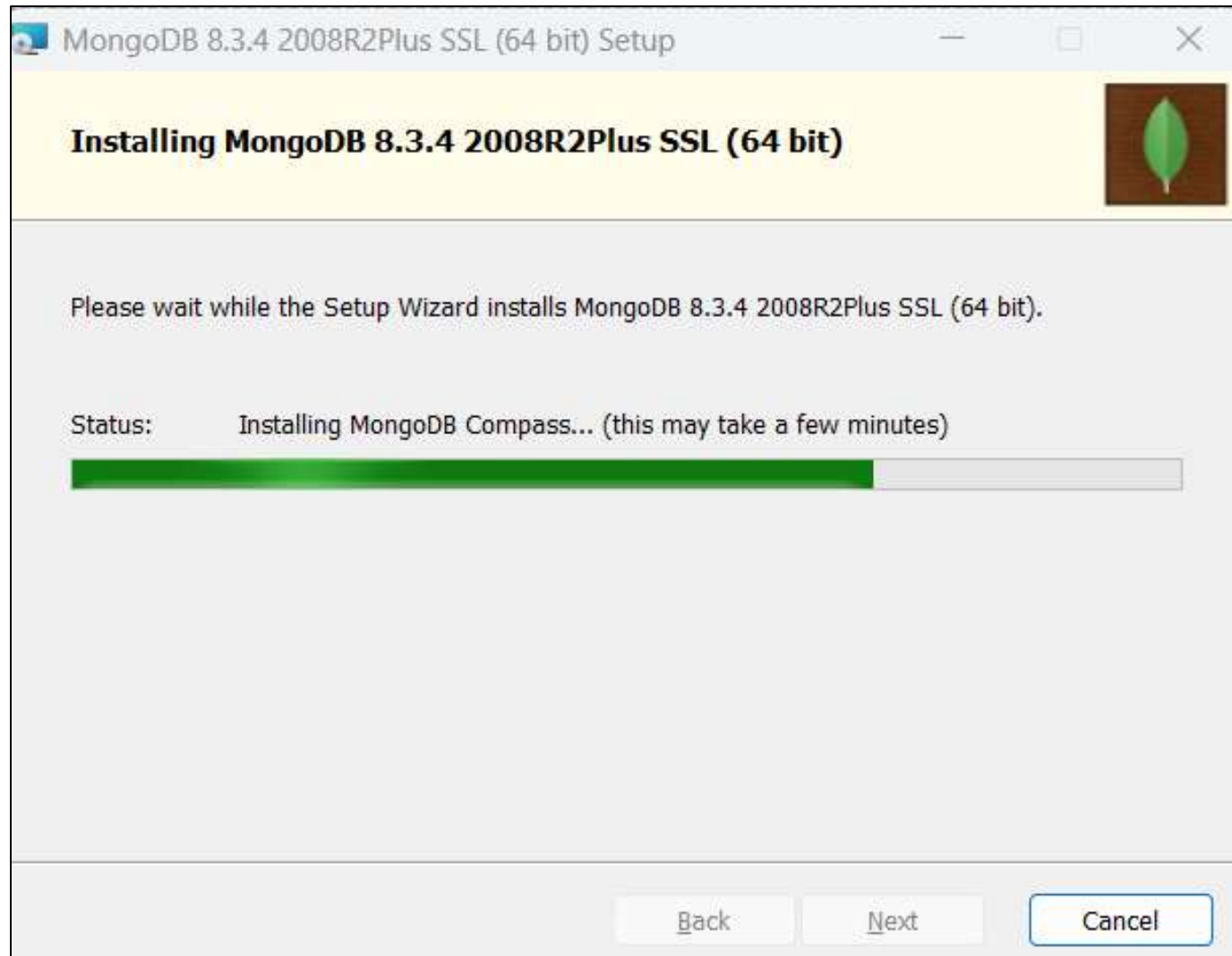
If the installer presents the Compass option and you want the GUI, keep **Install MongoDB Compass** selected.



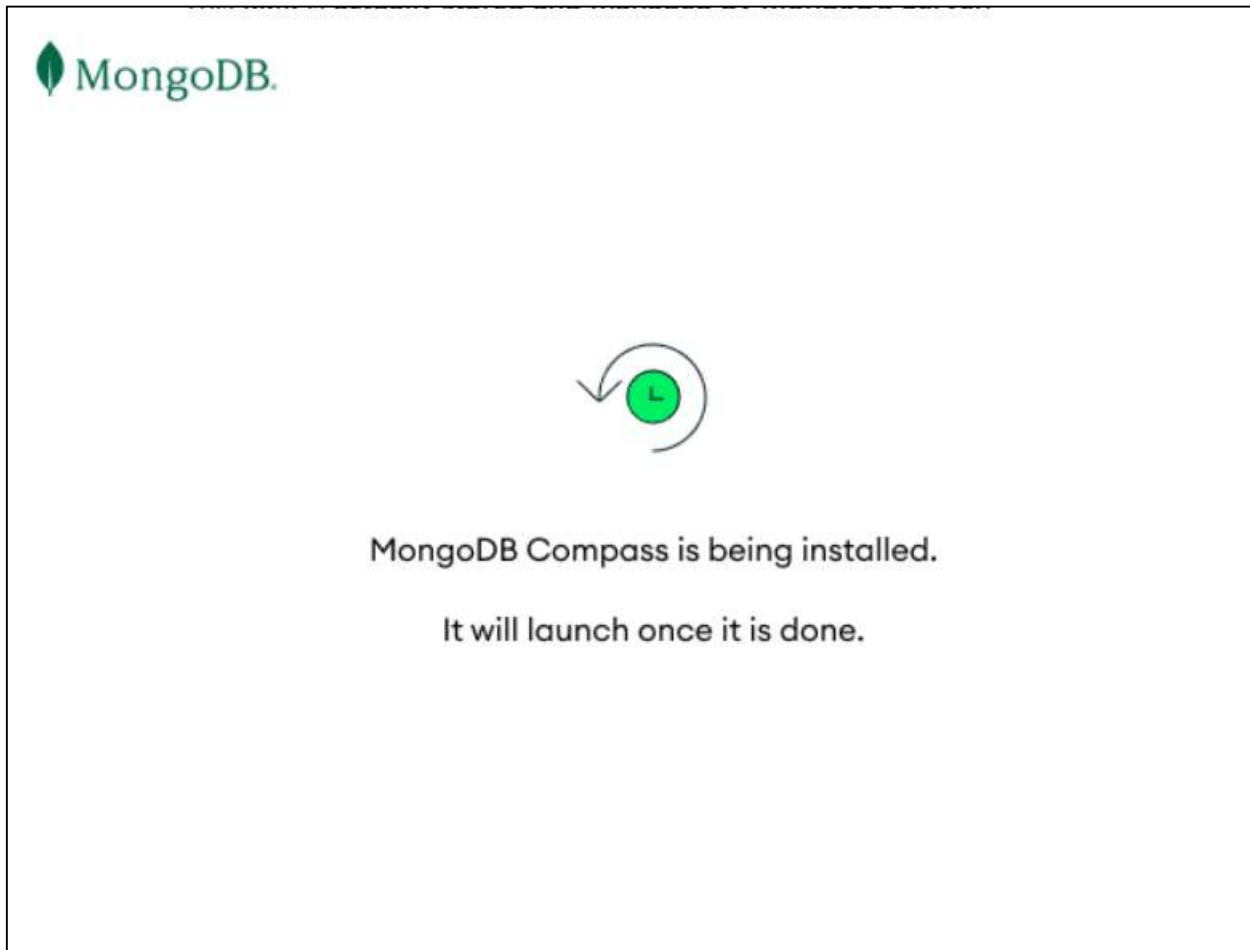
Click **Install** and complete the setup



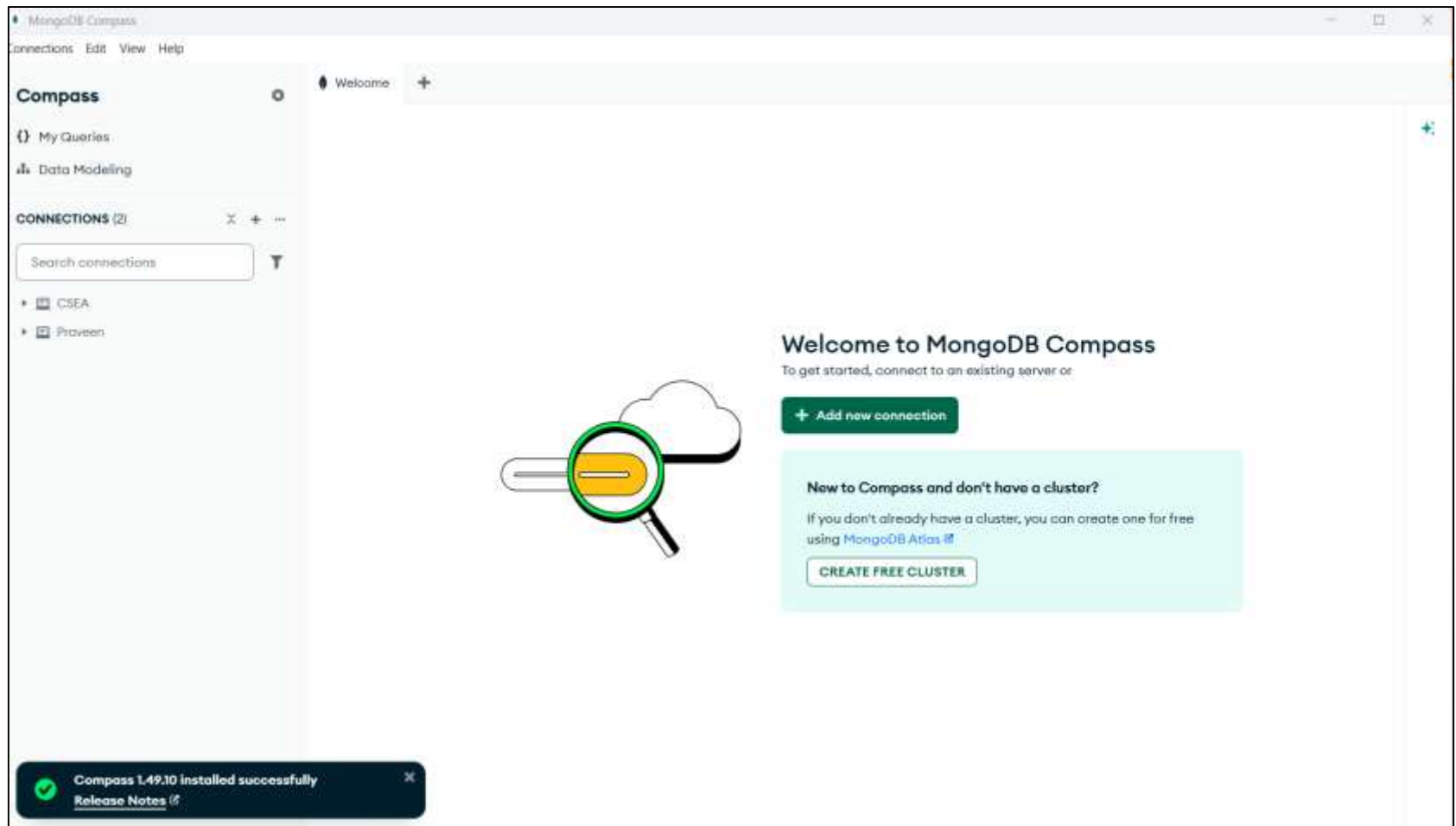
Installation is in progress....



Parallely MongoDB compass also installed

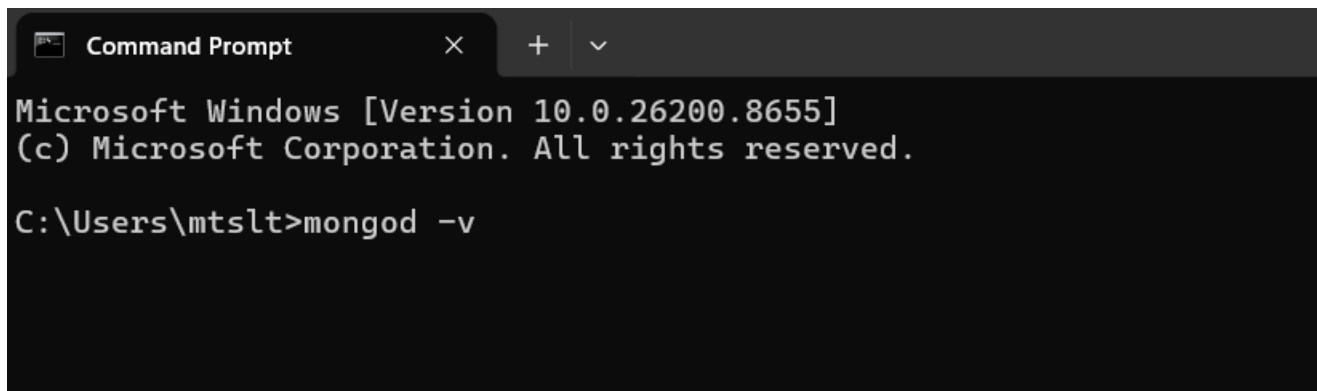


MongoDB Compass Installed Successfully



How to check Mongod Installed?

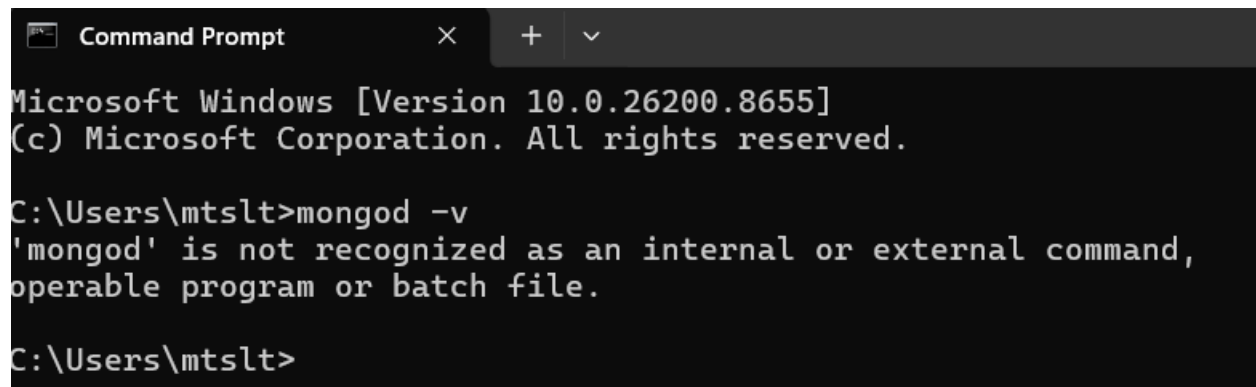
Open Cmd and Enter the command to check Mongod version



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mts\>mongod -v
```

If path is not setup, we get error as follows:



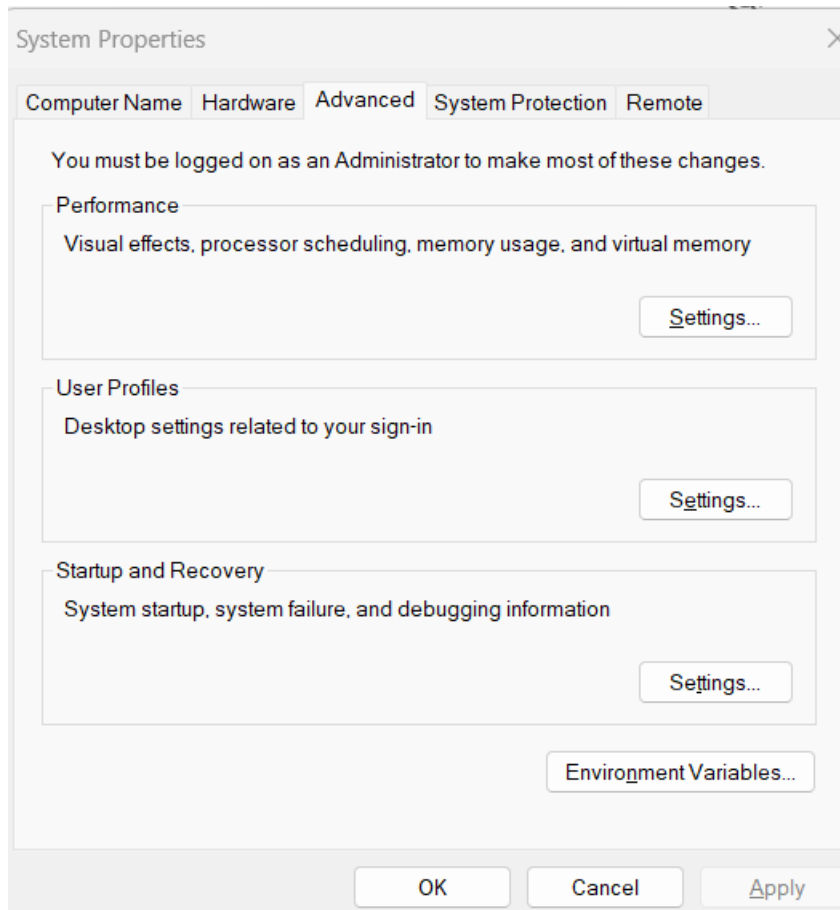
```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mts\>mongod -v
'mongod' is not recognized as an internal or external command,
operable program or batch file.

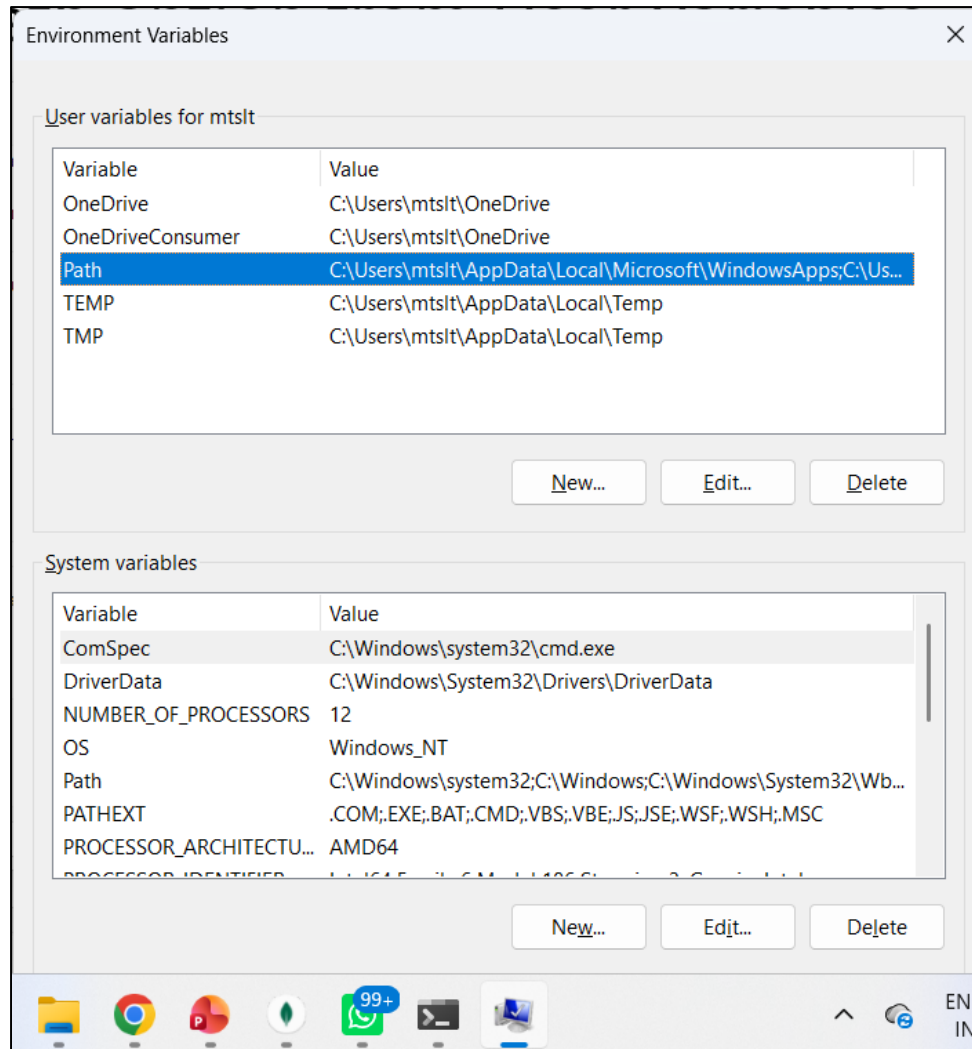
C:\Users\mts\>
```

How to setup Mongod Path ?

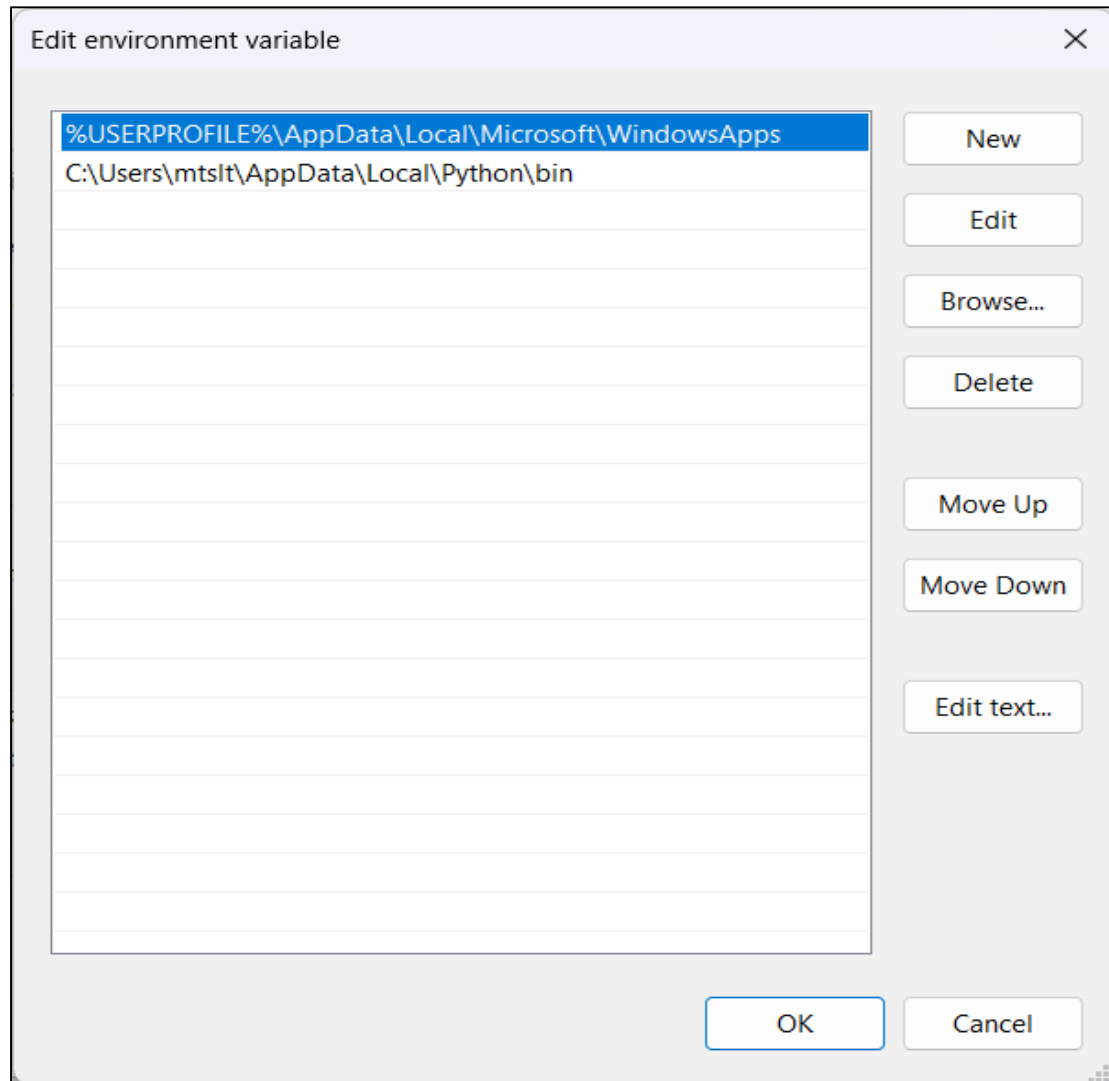
Open Environment Variables



Select Path option from User Variables and press edit button

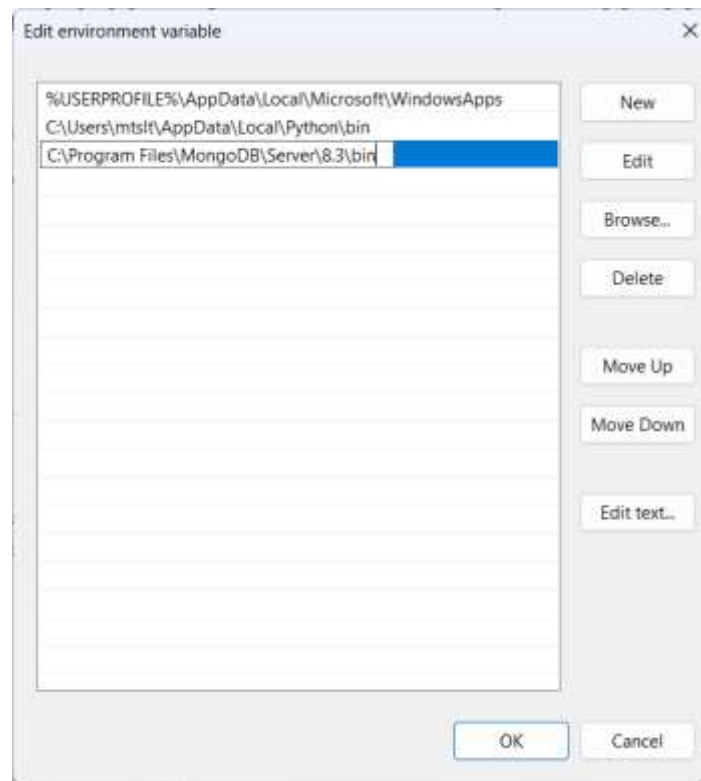


Click on New Button



Add the Corresponding Path using New Option

C:\Program Files\MongoDB\Server\8.3\bin



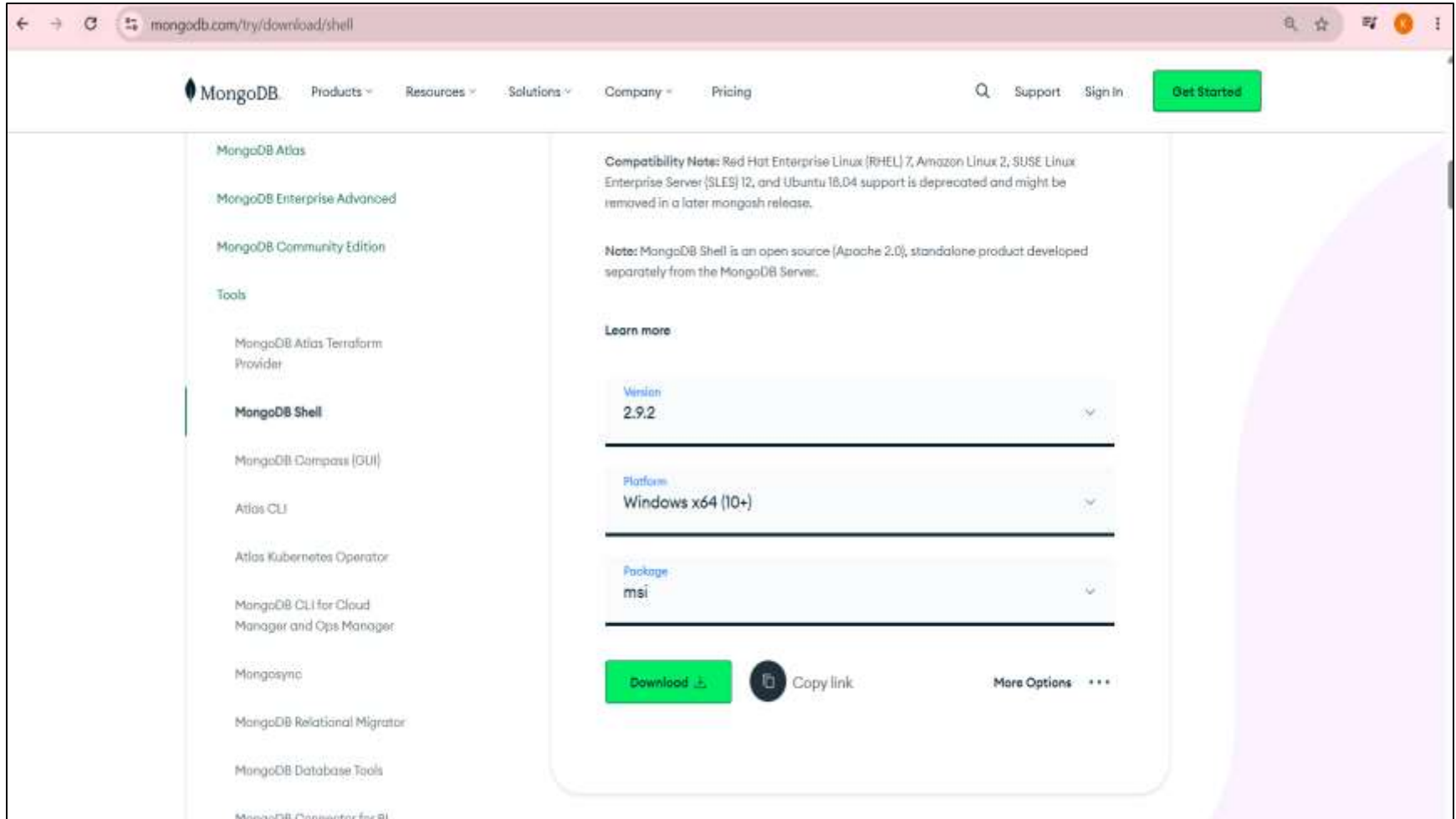
Press Ok Button after u copy that path

Now if we verify the same command in Command prompt

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mts\>mongod -v
{"t":{"$date":"2026-07-09T12:50:11.758+05:30"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1", "msg":"Shuffling
initializers", "attr":{"seed":1998911027}}
{"t":{"$date":"2026-07-09T12:50:11.774+05:30"},"s":"D1", "c":"CONTROL", "id":22615, "ctx":"thread1", "msg":"Loading li
brary: {toUtf8String_full_path_c_str}", "attr":{"toUtf8String_full_path_c_str":"Schannel.dll"}}
{"t":{"$date":"2026-07-09T12:50:11.778+05:30"},"s":"I", "c":"CONTROL", "id":97374, "ctx":"thread1", "msg":"Automatica
lly disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS
1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2026-07-09T12:50:11.843+05:30"},"s":"I", "c":"EXECUTOR", "id":11280000, "ctx":"thread1", "msg":"Starting t
hread pool", "attr":{"poolName":"AuthorizationRouter", "numThreads":0, "minThreads":0, "maxThreads":1000000000}}
{"t":{"$date":"2026-07-09T12:50:11.843+05:30"},"s":"I", "c":"EXECUTOR", "id":11280000, "ctx":"thread1", "msg":"Starting t
hread pool", "attr":{"poolName":"ThreadPool1", "numThreads":0, "minThreads":0, "maxThreads":1}}
{"t":{"$date":"2026-07-09T12:50:11.843+05:30"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"thread1", "msg":"Initialize
d wire specification", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0, "maxWireVersion":28}, "incomingInterna
lClient":{"minWireVersion":0, "maxWireVersion":28}, "outgoing":{"minWireVersion":6, "maxWireVersion":28}, "isInternalClient"
:true}}}}
{"t":{"$date":"2026-07-09T12:50:11.846+05:30"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1", "msg":"Multi thre
```

How to install MongoDB Shell



The screenshot shows the MongoDB website's download page for the MongoDB Shell. The browser address bar displays `mongodb.com/try/download/shell`. The navigation bar includes links for Products, Resources, Solutions, Company, and Pricing, along with a search icon, Support, Sign In, and a Get Started button.

The left sidebar lists various MongoDB products and tools, with **MongoDB Shell** highlighted. The main content area provides compatibility notes, a note about the shell being open source, and a 'Learn more' section. Below this, three dropdown menus allow users to select the version (2.9.2), platform (Windows x64 (10+)), and package type (msi). At the bottom, there are buttons for Download, Copy link, and More Options.

Compatibility Note: Red Hat Enterprise Linux (RHEL) 7, Amazon Linux 2, SUSE Linux Enterprise Server (SLES) 12, and Ubuntu 16.04 support is deprecated and might be removed in a later mongosh release.

Note: MongoDB Shell is an open source (Apache 2.0), standalone product developed separately from the MongoDB Server.

Learn more

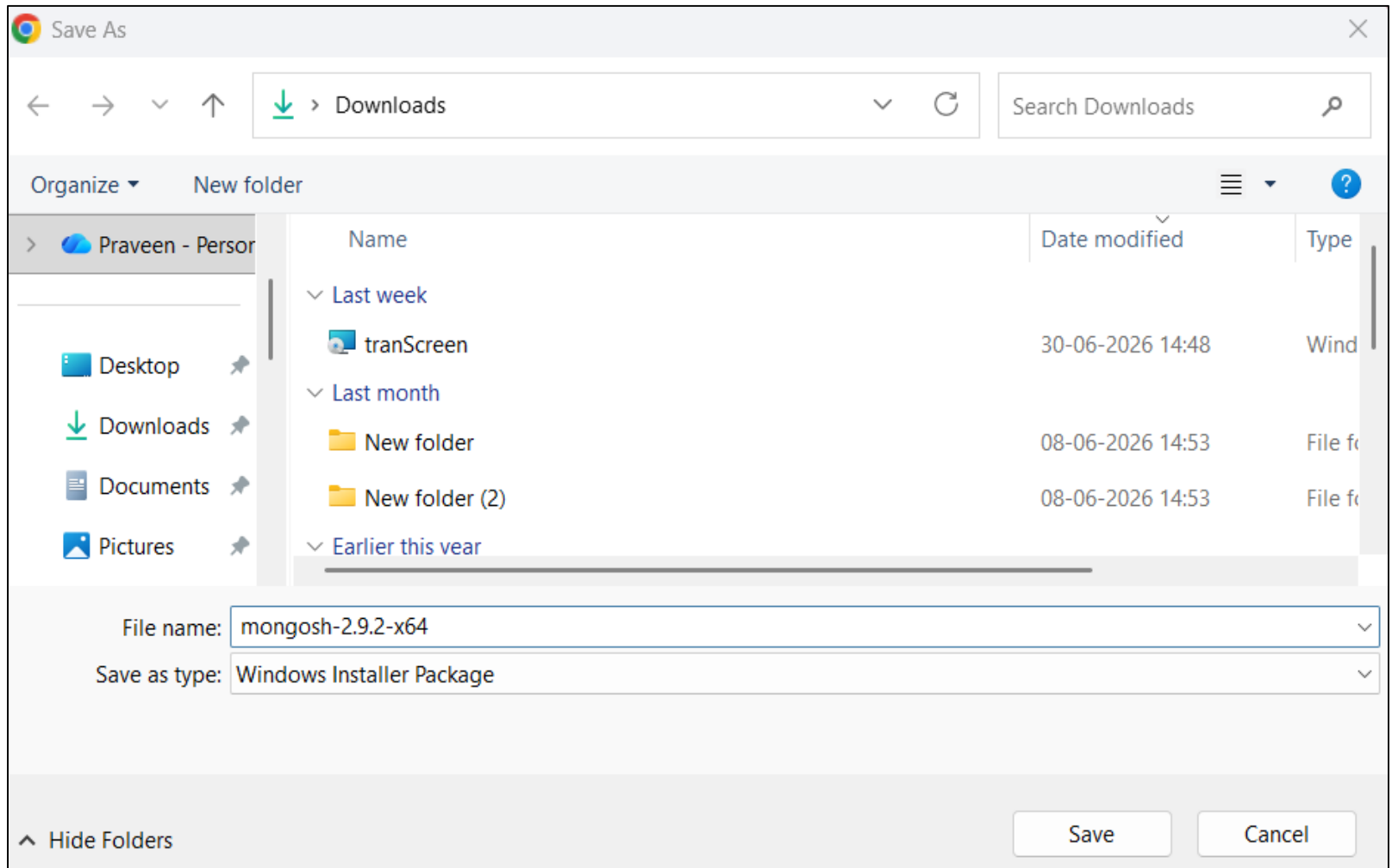
Version: 2.9.2

Platform: Windows x64 (10+)

Package: msi

[Download](#) [Copy link](#) [More Options](#)

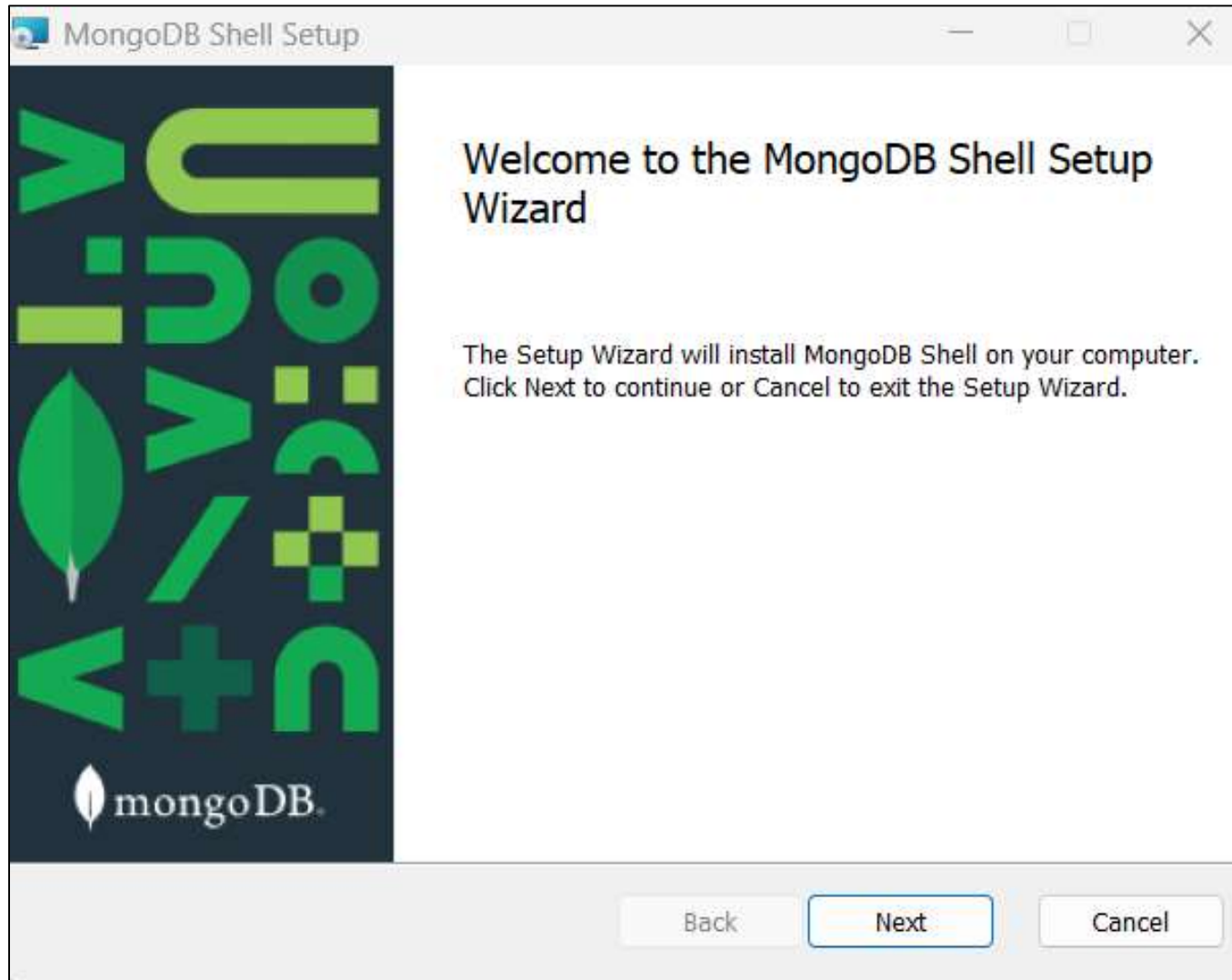
Click on download button



After downloading, Double click on
that Icon



Press Next Button



Press Next Button

MongoDB Shell Setup

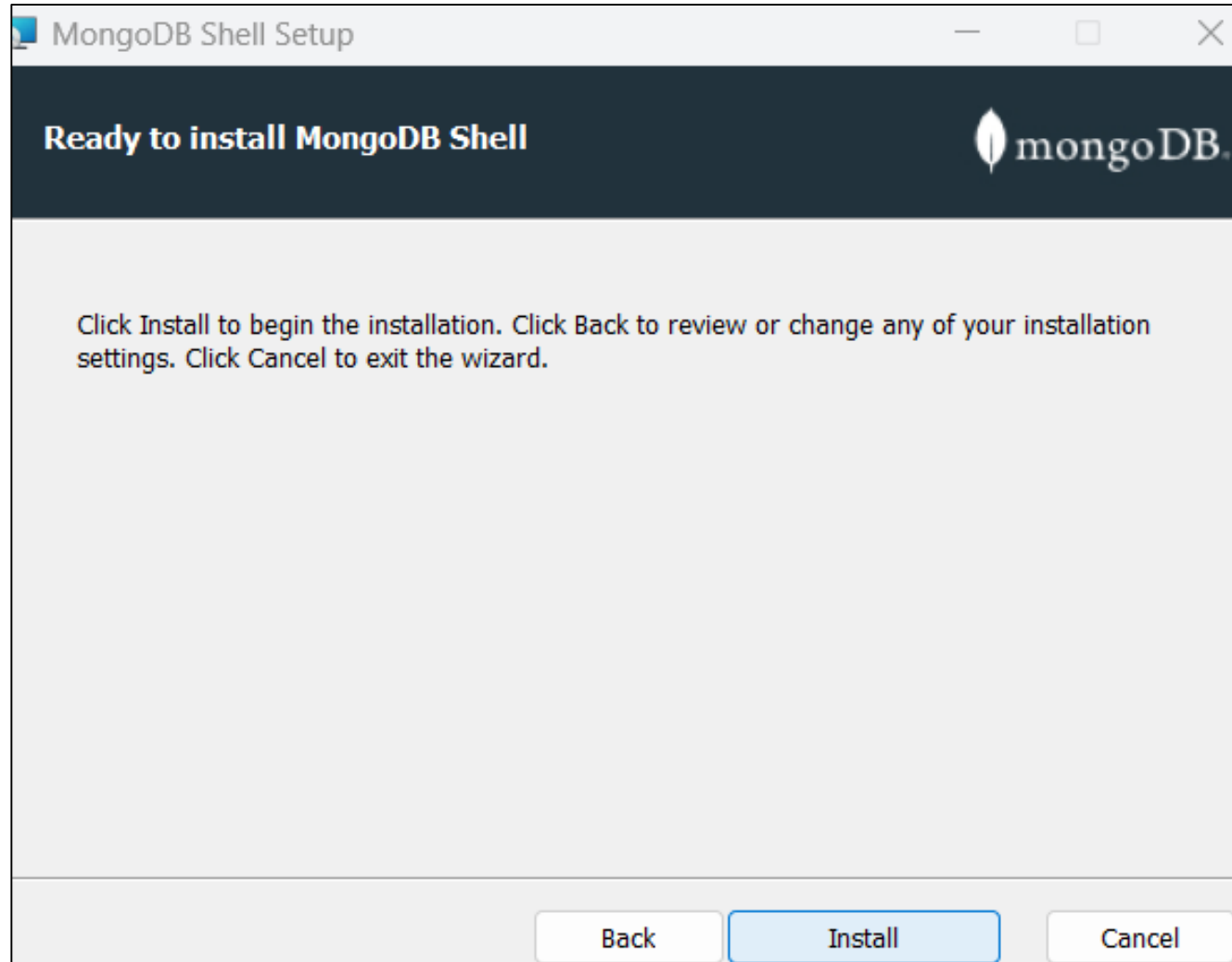
Destination Folder

Click Next to install to the default folder or click Change to choose another. 

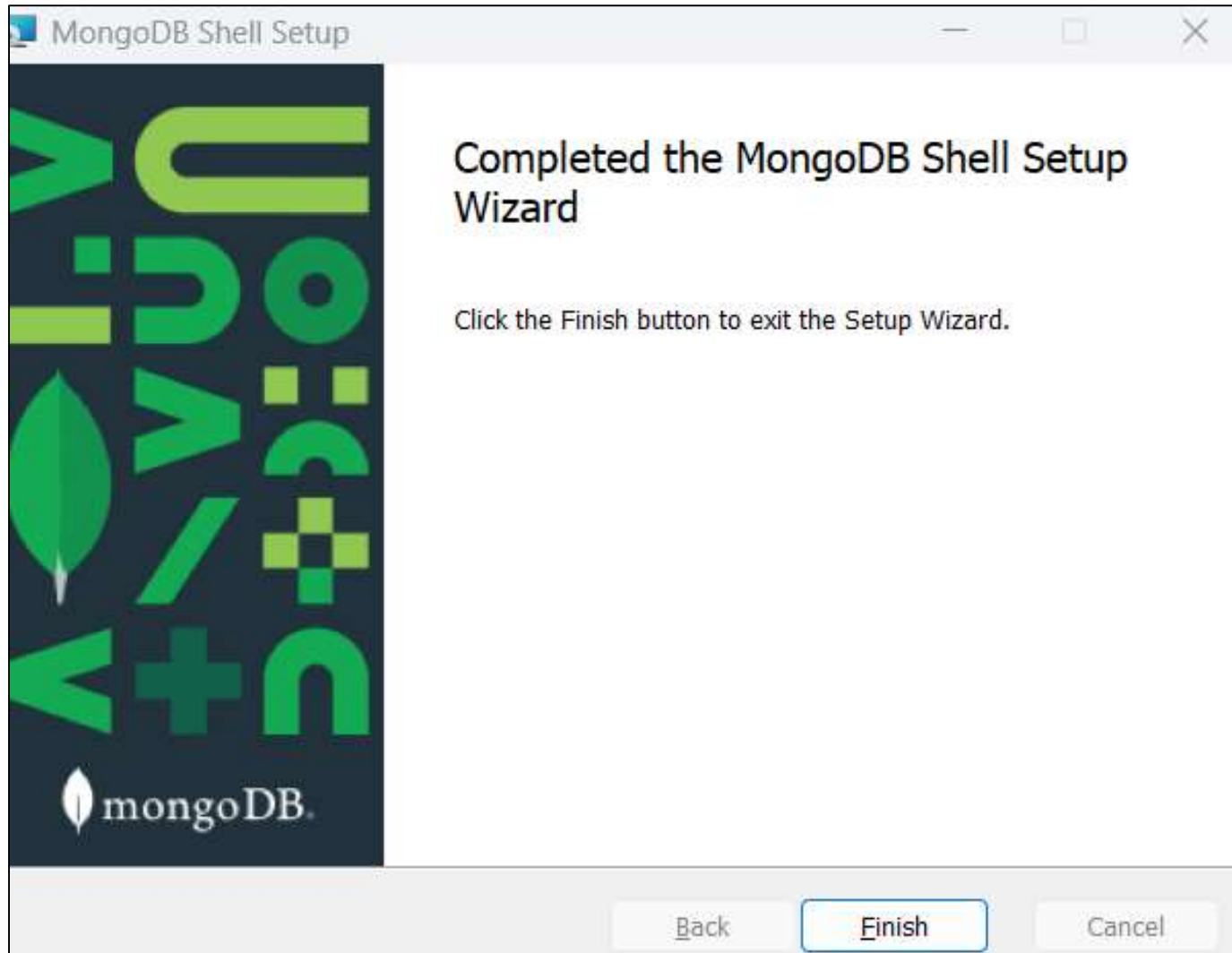
Install MongoDB Shell to:

☒ **Install just for you (mtsIt)**

Click Install Button

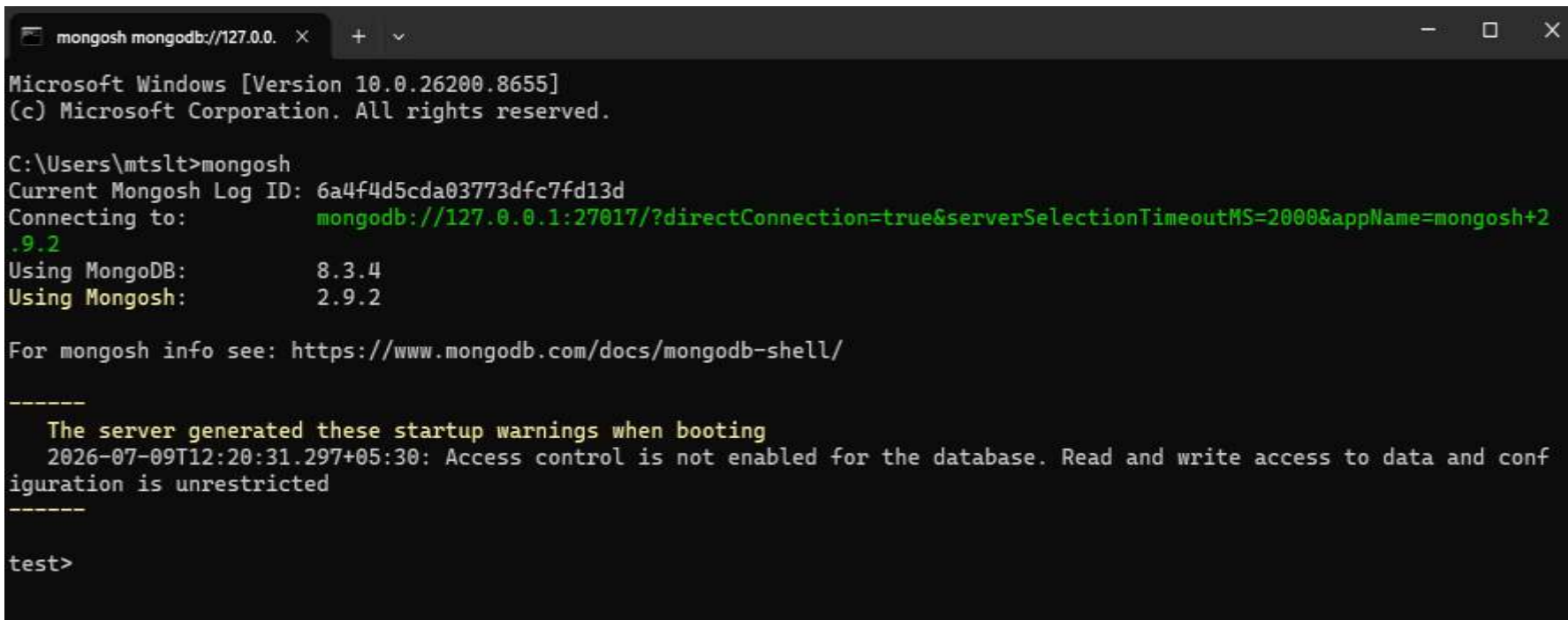


Click Finish Button



To verify mongosh installed or not?

Open cmd button and type command as
Cmd> mongosh



```
mongosh mongodb://127.0.0.1
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mts\>mongosh
Current Mongosh Log ID: 6a4f4d5cda03773dfc7fd13d
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.9.2
Using MongoDB:      8.3.4
Using Mongosh:       2.9.2

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-07-09T12:20:31.297+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test>
```